LITERATURE REVIEW

Adam: A Method for Stochastic Optimization

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ANALYTICAL ABSTRACT

The paper proposes Adam, a first-order method for minimising stochastic objectives that keeps exponential moving averages of the gradient and of the squared gradient, corrects both for their zero initialisation, and moves each parameter by the ratio of the two. The authors claim that the step is bounded by the learning rate, invariant to diagonal rescaling of the gradient, and well suited to sparse and non-stationary problems. They support the claim with a regret bound of order square root of T in online convex optimisation and with experiments on logistic regression, fully connected networks, convolutional networks and a variational autoencoder. This review reconstructs the update rule, the bias correction and the regret argument, evaluates the experiments, and tests the claims with an independent simulation. The empirical case for Adam as a robust default is persuasive, but the convergence theorem rests on a step in the proof that later work showed to be invalid, and the experiments cannot isolate which component of the algorithm produces the gains. The review ends with five research ideas, including applications to financial time series.

1 Introduction

1.1 Paper identification and classification

The paper is Kingma and Ba (2015), presented at ICLR 2015 and available as arXiv:1412.6980. It is a mixed paper: an algorithmic proposal, a theoretical regret analysis in the online convex setting, and an empirical evaluation on non-convex deep learning problems. The field is optimisation; the subfield is adaptive stochastic first-order methods. The version read in full is the arXiv text including the appendix and the reference list. The experiments are reported only as figures, so no numerical tables exist in the original.

1.2 Executive understanding

WHAT IS THE PAPER TRYING TO DISCOVER?

The paper looks for a stochastic gradient method that needs little tuning, uses little memory, behaves well when gradients are noisy, sparse or non-stationary, and still carries a convergence guarantee. The practical target is large neural networks trained on minibatches, where the objective is noisy and the gradient dimension is very high.

WHY DOES THE QUESTION MATTER?

Training deep models is dominated by the cost of gradient evaluations and of hyperparameter search. A method whose default settings work across architectures reduces both costs. For quantitative finance, the same property matters whenever neural models are estimated on short, noisy and non-stationary samples.

WHAT WAS KNOWN BEFORE, AND WHAT IS MISSING?

AdaGrad (Duchi, Hazan and Singer, 2011) adapts a per-coordinate step by the accumulated squared gradients and has strong regret guarantees for sparse data, but its step decays too aggressively when the gradient distribution is not stationary. RMSProp (Tieleman and Hinton, 2012) replaces the accumulation by an exponential average and works well in practice, but has no bias correction and no theory. Momentum methods (Sutskever et al., 2013) improve conditioning but do not adapt per coordinate. What was missing was a single method combining the three ideas with a proof.

WHAT DOES THE PAPER PROPOSE, HOW DOES IT INVESTIGATE, AND WHAT DOES IT FIND?

The paper proposes Adam, defined in Equations (1)–(4). It derives the properties of the effective step, proves a regret bound under convexity and bounded gradients, and compares Adam with SGD with Nesterov momentum, AdaGrad, RMSProp, AdaDelta and SFO on MNIST, IMDB, CIFAR-10 and a variational autoencoder. It finds that Adam is competitive with or better than the alternatives in

every reported experiment, and that omitting the bias correction destabilises training when the second-moment decay is close to one.

2 Research Problem and Hypothesis

2.1 Research question

PAPER Can combining a bias-corrected exponential average of the gradient with a bias-corrected exponential average of the squared gradient yield a per-coordinate adaptive step size that converges at the optimal online rate for convex problems and trains non-convex stochastic models at least as fast as the established alternatives, with one default hyperparameter setting?

2.2 Hypotheses

- The ratio of the first moment estimate to the root of the second moment estimate gives a step that is bounded by the learning rate and scale invariant, so no manual rescaling of gradients is needed.
- Bias correction of both moments is necessary for stability early in training, especially when the second-moment decay is close to one.
- Adam attains regret of order square root of T in online convex optimisation, with a data-dependent constant that is smaller when gradients are sparse.
- On real problems Adam matches or beats AdaGrad on sparse features, RMSProp and AdaDelta on dense networks, and SGD with momentum on convolutional networks.

2.3 Existing literature and research gap

PAPER The authors place Adam between AdaGrad and RMSProp. They show that AdaGrad is recovered when the first-moment decay is zero, the second-moment decay tends to one from below and the step size decays as $\alpha t^{-1/2}$, and that RMSProp differs by applying momentum to the rescaled gradient and by lacking bias correction.

CRITIQUE The gap is real in 2014: no method combined per-coordinate scaling, momentum and bias correction with an explicit regret statement. The gap is narrower than the paper suggests, because the theory covers a decaying step and decaying momentum, which are not the settings in which Adam is used or tested.

2.4 Claimed contribution

CRITIQUE Three contributions are claimed: the algorithm, the bias correction analysis and the regret bound. The algorithm and its empirical usefulness are demonstrated. The bias correction is demonstrated both analytically and in the variational autoencoder experiment. The regret bound is not demonstrated, since the proof contains an error identified by Reddi, Kale and Kumar (2018), discussed in Section 7.

3 Theoretical Framework and Mechanism

3.1 The update rule

PAPER Let $f(\theta)$ be a stochastic objective and $g_t = \nabla_\theta f_t(\theta_{t-1})$ the minibatch gradient at step t. Adam maintains

$$m_t = \beta_1 m_{t-1} + (1-\beta_1)\, g_t, \tag{1}$$

$$v_t = \beta_2 v_{t-1} + (1-\beta_2)\, g_t \odot g_t, \tag{2}$$

$$\hat{m}_t = \frac{m_t}{1-\beta_1^t}, \qquad \hat{v}_t = \frac{v_t}{1-\beta_2^t}, \tag{3}$$

$$\theta_t = \theta_{t-1} - \alpha \frac{\hat{m}_t}{\sqrt{\hat{v}_t} + \varepsilon}, \tag{4}$$

with $m_0 = v_0 = 0$ and defaults $\alpha = 0.001$, $\beta_1 = 0.9$, $\beta_2 = 0.999$, $\varepsilon = 10^{-8}$; all operations are element-wise.

g_t	Stochastic gradient at iteration t; a noisy estimate of the true gradient.
m_t	Exponential moving average of gradients with memory of about $1/(1-\beta_1) = 10$ steps.
v_t	Exponential moving average of squared gradients, memory of about $1/(1-\beta_2) = 1000$ steps; an estimate of the uncentred second moment.
$\hat{m}_t, \hat{v}_t$	Bias-corrected estimates, removing the pull towards the zero initial value.
α	Step size, the maximum scale of a parameter change per iteration.
ε	Numerical floor that prevents division by zero when $\hat{v}_t$ is very small.

Equation (4) divides a smoothed gradient by a smoothed gradient magnitude. A coordinate with consistently signed gradients has $|\hat{m}_t| \approx \sqrt{\hat{v}_t}$ and moves by about α per step. A coordinate whose gradient changes sign has $\hat{m}_t$ close to zero relative to $\sqrt{\hat{v}_t}$ and moves little. The ratio $\hat{m}_t/\sqrt{\hat{v}_t}$ is therefore a signal-to-noise ratio, which the authors use to explain automatic annealing near an optimum. The method is a smoothed, normalised relative of sign descent: when $\beta_1 = \beta_2 = 0$ and $\varepsilon = 0$, Equation (4) reduces to $\theta_t = \theta_{t-1} - \alpha\,\mathrm{sign}(g_t)$.

3.2 Bounds on the effective step and scale invariance

PAPER With $\varepsilon = 0$ the effective step is $\Delta_t = \alpha\, \hat{m}_t / \sqrt{\hat{v}_t}$. The paper states that

$$|\Delta_t| \le \alpha \frac{1-\beta_1}{\sqrt{1-\beta_2}} \text{ if } (1-\beta_1) > \sqrt{1-\beta_2}, \qquad |\Delta_t| \le \alpha \text{ otherwise.} \tag{5}$$

The first case occurs only in the sparsest setting, when the gradient is zero at all previous steps except the current one. With the defaults, $(1-\beta_1)/\sqrt{1-\beta_2} = 0.1/0.0316 = 3.16$, so the largest possible step is about $3.16\,\alpha$ and in non-degenerate cases it is below α. Replacing g_t by $c\,g_t$ multiplies $\hat{m}_t$ by c and $\sqrt{\hat{v}_t}$ by $|c|$, so Δ_t is unchanged when $\varepsilon = 0$. **INFERENCE** The invariance is exact only without ε; with $\varepsilon = 10^{-8}$ it fails for gradients of order 10^{-8} or smaller, which arise in deep networks with vanishing gradients (see Section 8).

3.3 Initialisation bias correction

PAPER Unrolling Equation (2) from $v_0 = 0$ gives

$$v_t = (1-\beta_2) \sum_{i=1}^{t} \beta_2^{t-i}\, g_i \odot g_i. \tag{6}$$

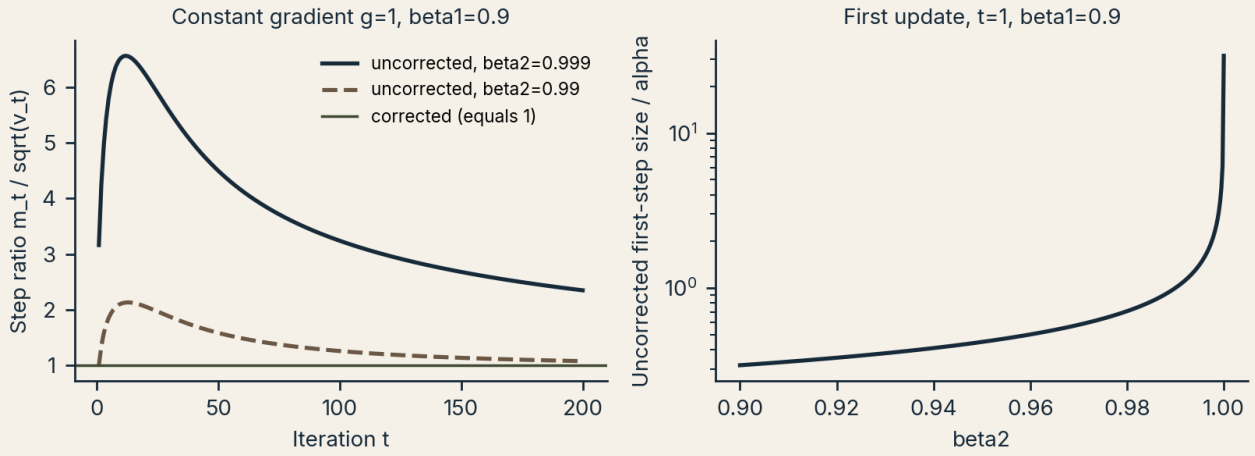
Taking expectations, and writing ζ for the error from non-stationarity,

$$\mathbb{E}[v_t] = \mathbb{E}[g_t^2]\,(1-\beta_2^t) + \zeta. \tag{7}$$

$\mathbb{E}[g_t^2]$	True second moment of the gradient at step t.
$1-\beta_2^t$	Sum of the geometric weights $(1-\beta_2)\sum_{i=1}^{t}\beta_2^{t-i}$, equal to the total weight placed on the observed gradients.
ζ	Residual, zero if the gradient second moment is stationary, and small if β_2 gives little weight to old gradients.

The geometric weights sum to $1-\beta_2^t$ rather than to one, so v_t understates the second moment by that factor; dividing by it yields $\hat{v}_t$. The effect is large early: at $t = 1$ with $\beta_2 = 0.999$, $v_1 = 0.001\, g_1^2$, so without correction the step would be $\hat{m}_1/\sqrt{v_1}$ with $\hat{m}_1 = g_1$ and $\sqrt{v_1} = 0.0316\,|g_1|$, thirty-two times too large. Dropping only the second correction, with the first-moment correction kept, makes the first step $\hat{m}_1/\sqrt{v_1}$ equal to $1/0.0316 = 31.6$ in magnitude; dropping both gives $0.1/0.0316 = 3.16$, the figure used in Equation (5). Figure 1 plots the uncorrected ratio under a constant gradient.

Figure 1. **Effect of omitting bias correction under a constant gradient** $g_t = 1$



Source: generated by the reviewer from Equations (1) and (2) with $m_0 = v_0 = 0$, $\beta_1 = 0.9$; no data used. Left: ratio $m_t/\sqrt{v_t}$ without correction; the corrected ratio equals one at every step. Right: size of the uncorrected first step as a multiple of α.

EVIDENCE The left panel shows that the uncorrected ratio is far from one for several hundred steps when $\beta_2 = 0.999$, and the right panel shows that the first step grows without bound as $\beta_2 \to 1$. This is precisely the regime in which the paper sets its default. The correction is therefore not cosmetic: it makes the default hyperparameters usable.

3.4 The regret framework and Theorem 4.1

PAPER The analysis follows Zinkevich (2003). At each round an adversary reveals a convex function f_t and the algorithm has already chosen θ_t in a convex compact set $\mathcal{X}$. Regret is

$$R(T) = \sum_{t=1}^{T} \left[f_t(\theta_t) - f_t(\theta^*) \right], \qquad \theta^* = \arg\min_{\theta \in \mathcal{X}} \sum_{t=1}^{T} f_t(\theta). \tag{8}$$

Assumptions are $\|\nabla f_t(\theta)\|_2 \le G$, $\|\nabla f_t(\theta)\|_\infty \le G_\infty$, $\|\theta_n - \theta_m\|_2 \le D$, $\|\theta_n - \theta_m\|_\infty \le D_\infty$, $\gamma = \beta_1/\sqrt{\beta_2} < 1$, a decaying step $\alpha_t = \alpha/\sqrt{t}$ and decaying momentum $\beta_{1,t} = \beta_1 \lambda^{t-1}$ with $\lambda \in (0, 1)$. Theorem 4.1 states

$$R(T) \le \frac{D^2}{2\alpha(1-\beta_1)} \sum_{i=1}^{d} \sqrt{T \hat{v}_{T,i}} + \frac{\alpha(1+\beta_1) G_\infty}{(1-\beta_1)\sqrt{1-\beta_2}\,(1-\gamma)^2} \sum_{i=1}^{d} \|g_{1:T,i}\|_2 + \sum_{i=1}^{d} \frac{D_\infty^2\, G_\infty \sqrt{1-\beta_2}}{2\alpha(1-\beta_1)(1-\lambda)^2}. \tag{9}$$

$g_{1:T,i}$	Vector of the i-th coordinate of the gradients from step 1 to T.
$\hat{v}_{T,i}$	Final bias-corrected second moment of coordinate i.
d	Dimension of the parameter vector.

Corollary 4.2 divides by T and concludes $R(T)/T = O(1/\sqrt{T})$. The first term is $O(\sqrt{T})$ because $\hat{v}_{T,i}$ is bounded by G_∞^2. The second term is $O(\sqrt{T})$ through Lemma 10.3 of the paper, which states

$\sum_{t=1}^{T} \sqrt{g_{t,i}^2/t} \le 2G_\infty \|g_{1:T,i}\|_2$. The third term is a constant that comes from summing the momentum decay.

INFERENCE Reconstruction of the proof. Let Π denote the projection on $\mathcal{X}$ in the metric of the diagonal matrix $\sqrt{\hat{v}_t}$. Writing the update as $\theta_{t+1} = \Pi(\theta_t - \alpha_t \hat{m}_t/\sqrt{\hat{v}_t})$ and using convexity, $f_t(\theta_t) - f_t(\theta^*) \le g_t^\top(\theta_t - \theta^*)$. Substituting $g_t = (m_t - \beta_{1,t} m_{t-1})/(1-\beta_{1,t})$ and expanding the squared distance gives, for each coordinate i,

$$g_{t,i}(\theta_{t,i} - \theta_i^*) \le \frac{\sqrt{\hat{v}_{t,i}}}{2\alpha_t(1-\beta_{1,t})}\Big[(\theta_{t,i} - \theta_i^*)^2 - (\theta_{t+1,i} - \theta_i^*)^2\Big] + (\text{momentum cross term}) + \frac{\alpha_t \hat{m}_{t,i}^2}{2(1-\beta_{1,t})\sqrt{\hat{v}_{t,i}}}. \quad (10)$$

Summing over t, the bracketed term is meant to telescope. After telescoping, its contribution equals

$$\frac{1}{2(1-\beta_1)} \sum_t \Big(\frac{\sqrt{\hat{v}_{t,i}}}{\alpha_t} - \frac{\sqrt{\hat{v}_{t-1,i}}}{\alpha_{t-1}} \Big) (\theta_{t,i} - \theta_i^*)^2, \quad (11)$$

which is bounded by $D_\infty^2 \sqrt{T \hat{v}_{T,i}}/(2\alpha(1-\beta_1))$ *only if each weight difference is non-negative*, that is, only if $\sqrt{\hat{v}_{t,i}}/\alpha_t$ is non-decreasing in t. The third term of (10) is controlled by Lemmas 10.3 and 10.4 using $\gamma < 1$; the cross term is controlled by Young's inequality and the geometric decay $\beta_{1,t} = \beta_1 \lambda^{t-1}$, producing the constant in the third line of (9).

CRITIQUE The non-decreasing condition is the flaw. With $\alpha_t = \alpha/\sqrt{t}$ and $\hat{v}_t$ computed from an exponential average, $\hat{v}_{t,i}$ can fall when a large gradient leaves the averaging window faster than α_t falls. When the difference is negative, the telescoped bound no longer holds. Reddi, Kale and Kumar (2018) identified this step, and their counterexample in Section 8 shows that the theorem as stated is false. The reviewer also notes, as read from the text, that the appendix restatement of the theorem differs from the main-text version in the denominator of the last term; this should be checked against the published proceedings.

INFERENCE With the default constants the bound is also very loose. With $\beta_1 = 0.9$ and $\beta_2 = 0.999$, $\gamma = 0.9/\sqrt{0.999} = 0.9005$, so $(1-\gamma)^{-2} \approx 101$ and the coefficient of the second term is about $1.9/(0.1 \times 0.0316 \times 0.0099) \approx 6 \times 10^4$ times αG_∞ per unit of $\sum_i \|g_{1:T,i}\|_2$. Reviewer computation. A bound with such a constant says little for the horizons used in practice.

3.5 Relations to AdaGrad, RMSProp and the AdaMax extension

PAPER Setting $\beta_1 = 0$, letting $\beta_2 \to 1^-$ and $\alpha_t = \alpha t^{-1/2}$ gives

$$\theta_{t+1} = \theta_t - \alpha \frac{g_t}{\sqrt{\sum_{i=1}^t g_i^2}}, \quad (12)$$

which is AdaGrad. The equivalence needs the bias correction: $\lim_{\beta_2 \to 1} \hat{v}_t = \frac{1}{t}\sum_{i \le t} g_i^2$, and $\alpha_t / \sqrt{\hat{v}_t} = \alpha / \sqrt{\sum g_i^2}$. The AdaMax variant replaces the L^2 norm by the L^∞ norm,

$$u_t = \max(\beta_2 u_{t-1}, |g_t|), \qquad \theta_t = \theta_{t-1} - \frac{\alpha}{1-\beta_1^t}\frac{m_t}{u_t}, \tag{13}$$

with defaults $\alpha = 0.002$. Because $u_t \ge |g_t|$ is a running maximum, no bias correction is needed for u_t and $|\Delta_t| \le \alpha$ holds exactly. The paper also proposes an exponential average of the iterates, $\bar{\theta}_t = \beta_2 \bar{\theta}_{t-1} + (1-\beta_2)\theta_t$ with correction $\bar{\theta}_t / (1-\beta_2^t)$.

4 Data and Empirical Strategy

4.1 Data

The paper uses standard benchmarks, summarised in Table 1. No train, validation and test split is described in the text read, and the figures plot training cost, not test error, in the main comparisons.

Table 1. **Experiments reported in the paper**

Experiment	Model and data	Comparators	Reported outcome (figures only)
Logistic regression, dense	L2-regularised multiclass, MNIST, minibatch 128, $\alpha/\sqrt{t}$	SGD with Nesterov momentum, AdaGrad	Adam matches SGD with momentum and beats AdaGrad
Logistic regression, sparse	IMDB bag-of-words, 10,000 dimensions, 50 percent input dropout	SGD with Nesterov, AdaGrad	AdaGrad beats SGD; Adam converges as fast as AdaGrad
Multilayer network	Two hidden layers of 1000 ReLU units, MNIST, minibatch 128	SFO, then AdaGrad, RMSProp, SGD Nesterov, AdaDelta with dropout	Adam faster than SFO per iteration and in wall-clock time; best with dropout
Convolutional network	CIFAR-10, c64-c64-c128 then 1000 ReLU units, 45 epochs	AdaGrad, SGD with momentum	Adam and SGD converge faster than AdaGrad; small gain over SGD with momentum
Bias correction	Variational autoencoder, 500 softplus units, 50 latent dimensions	Adam without correction, RMSProp	Without correction, unstable when β_2 near one; with correction, equal or better than RMSProp

Source: reviewer's summary of the experiments as reported in Kingma and Ba (2015). Outcomes are qualitative statements read from the paper's figures (official).

4.2 Empirical specification and identification

CRITIQUE The experiments are a comparison of optimisers on fixed models, so identification here means attributing the speed of convergence to the optimiser rather than to hyperparameters, initialisation or random seeds. The identifying variation is the choice of optimiser with the architecture and data held constant, and, in the bias correction study, a sweep over $\beta_1 \in [0, 0.9]$, $\beta_2 \in \{0.99, 0.999, 0.9999\}$ and $\log_{10}\alpha \in [-5, -1]$. The paper reports that learning rates were tuned for each method over a grid; the width of that grid for competitors is not described in the text read, which is the main threat to comparability. The experiments cannot separate the contributions of momentum, per-coordinate scaling and bias correction, since no ablation holds two components fixed while varying the third, apart from the bias correction sweep.

4.3 Data limitations and biases

- Training cost is the main outcome. Faster minimisation of training cost does not imply better generalisation, which is the quantity of interest for statistical models.
- Tuning budgets are unequal by construction: Adam has stable defaults while competitors need tuning, and the paper does not report the number of configurations tried per method.
- Results appear to come from single runs or from few seeds; the figures show no confidence bands, so differences of a few percent cannot be interpreted.
- All benchmarks are small image or text tasks of 2014; heavy-tailed, non-stationary and regression-type targets, typical of financial data, are not covered.

5 Results

5.1 Logistic regression and multilayer networks

EVIDENCE On MNIST logistic regression, Adam tracks SGD with Nesterov momentum and is faster than AdaGrad in the early epochs. On IMDB the sparse input favours the adaptive methods, and Adam matches AdaGrad, which is consistent with the claim that Adam inherits AdaGrad's advantage for sparse features. On the multilayer network, Adam beats SFO in iterations and in wall-clock time, even though SFO uses curvature information; the paper reports that SFO costs five to ten times more per iteration and does not converge with dropout. With dropout, Adam is reported best among AdaGrad, RMSProp, SGD Nesterov and AdaDelta. The magnitudes are shown only graphically, so no coefficient or effect size can be reported. CRITIQUE The qualitative ordering is convincing across settings; its economic size, such as wall-clock time to a target loss, is not quantified.

5.2 Convolutional networks

EVIDENCE On CIFAR-10, AdaGrad falls quickly at first but is slower over 45 epochs; Adam and SGD with momentum converge faster, with only a marginal advantage for Adam. The authors attribute AdaGrad's slowdown to $\hat{v}_t$ becoming dominated by ε in the first layers. INFERENCE That attribution is plausible but untested: a direct measurement of the share of coordinates with $\sqrt{\hat{v}_t} < \varepsilon$ would settle it.

5.3 Bias correction

EVIDENCE In the variational autoencoder sweep, runs without correction diverge or stall when β_2 is close to one, in line with Figure 1; with correction the method performs equal to or better than RMSProp over the grid. This is the cleanest experiment of the paper, because a single component is switched off while everything else is fixed.

5.4 What the regret bound delivers numerically

INFERENCE Table 2 evaluates the coefficients of Equation (9) under the defaults, to translate the theorem into magnitudes.

Table 2. **Coefficients of Theorem 4.1 under the default hyperparameters**

Quantity	Value	Comment
$\gamma = \beta_1/\sqrt{\beta_2}$	0.9005	Must be below one
$(1-\gamma)^{-2}$	about 101	Amplifies the second term
$(1-\beta_1)/\sqrt{1-\beta_2}$	3.16	Maximum step multiple, sparse case
Coefficient of $\alpha G_\infty \sum_i \lVert g_{1:T,i} \rVert_2$	about 6×10^4	Product of the factors in the second term

Source: computed by the reviewer from the formulas of Theorem 4.1 (generated; not reported in the paper).

> **READING OF THE TABLE**
>
> The theorem is asymptotically meaningful, since the average regret falls as one over the square root of T, but with the recommended defaults the constants are so large that the bound is uninformative for horizons below several million iterations. The sparsity-dependent sum $\sum_i \lVert g_{1:T,i} \rVert_2$ is the only place where the data enter.

6 Robustness and Additional Tests

Table 3. **Robustness tests: concern, test, result and assessment**

Concern	Test reported	Result	Does it resolve the concern?
Is bias correction needed?	VAE sweep over β_1, β_2, α with and without correction	Instability without correction when β_2 near one	Yes, for this model; one architecture only
Does Adam depend on the architecture?	Logistic regression, MLP, CNN, VAE	Competitive in all	Partly; four small models, no transformers, recurrent nets or regression targets
Does Adam depend on regularisation?	MLP with and without dropout; deterministic cost with weight decay	Better than SFO and RMSProp in both	Partly; only two regularisers
Sparse versus dense features	IMDB bag-of-words versus MNIST	Matches AdaGrad on sparse, SGD on dense	Partly; no direct measurement of gradient sparsity
Are the competitors fairly tuned?	Learning rates tuned over a grid (as stated)	Adam ahead of or level with the best	No; grids and seeds not reported, so tuning asymmetry cannot be excluded
Non-convex guarantee	None; conclusion states the theory covers the convex case	Not applicable	No; the practical use is entirely outside the theorem

Source: reviewer's assessment of Kingma and Ba (2015).

7 Critical Evaluation

7.1 What the paper establishes

ESTABLISHED

The update rule is a bounded, scale-invariant (for $\varepsilon = 0$) normalised momentum step, with the bounds in Equation (5). Bias correction is required for stable early training when β_2 is near one. Adam trains the reported small and medium models at least as fast as the compared methods under default settings.

SUPPORTED BUT CONDITIONAL

Adam's advantage on sparse features, which rests on one text dataset and on comparison with AdaGrad. The claim that default hyperparameters transfer, which holds across the four reported models only.

> **SUGGESTIVE**
>
> The interpretation of $\hat{m}_t/\sqrt{\hat{v}_t}$ as a signal-to-noise ratio that anneals the step automatically. The statement that $\hat{v}_t$ approximates the diagonal of the Fisher information in the sense of natural gradient. Neither is tested.

> **NOT ESTABLISHED**
>
> The regret bound of Theorem 4.1 as stated. Convergence in non-convex problems. Any claim about generalisation. Any claim that each of the three components is individually necessary.

7.2 Strengths

STRENGTHS RECOGNISED BY THE AUTHORS

- Low memory and compute: two extra vectors of parameter size. This matters because it makes the method usable for the largest models.
- Invariance to diagonal gradient rescaling and bounded steps, which reduce the sensitivity to the units of the loss.
- Suitability to sparse and non-stationary gradients, as shown by the IMDB and dropout experiments.

STRENGTHS IDENTIFIED INDEPENDENTLY

- Practical impact: Adam became the default optimiser in most deep learning libraries, which is evidence of external validity beyond the original experiments.
- The bias correction is a small, correct and transferable idea, easy to verify analytically as in Equations (6) and (7).
- The connection to AdaGrad in Equation (12) unifies several methods and led to later families such as AdaMax and AMSGrad.
- The paper is short and its algorithm is unambiguous, which makes replication easy, as the simulation below illustrates.

7.3 Weaknesses and critique

SINGLE MOST IMPORTANT WEAKNESS

Problem. The convergence theorem, the paper's only formal result, relies on a telescoping step that requires $\sqrt{\hat{v}_{t,i}}/\alpha_t$ to be non-decreasing, which the exponential moving average does not guarantee. **Why it matters.** The theorem is the stated justification for using an exponential average instead of AdaGrad's accumulation, and it is false in general: there exist convex problems on which Adam does not converge to the optimal average regret. **Potential bias.** Users read the theorem as a guarantee

and tune β_2 without recognising the risk in problems with rare but large gradients. **How it could be tested.** Run Adam on the Reddi, Kale and Kumar construction and compare with a version that keeps the running maximum of $\hat{v}_t$, as done in Section 8.

OTHER WEAKNESSES

- Gap between theory and practice. The proof needs a decaying step $\alpha/\sqrt{t}$ and decaying $\beta_{1,t}$, while the experiments use constant β_1 and, in most cases, a constant step. The bound therefore does not apply to the algorithm that is evaluated. Test: prove or simulate the constant-parameter case.
- No ablation of momentum, scaling and correction. Without it, the benefit cannot be attributed. Test: factorial experiment over the three switches.
- Training cost instead of generalisation. Adaptive methods can reach lower training loss and worse test error, as later documented by Wilson et al. (2017). Test: report validation error at matched training loss.
- Choice of the stopping and tuning budget is unreported, creating a possible multiple-testing advantage for the proposed method.
- The role of ε is neglected: it is a hyperparameter in practice, and it breaks scale invariance (Section 8).

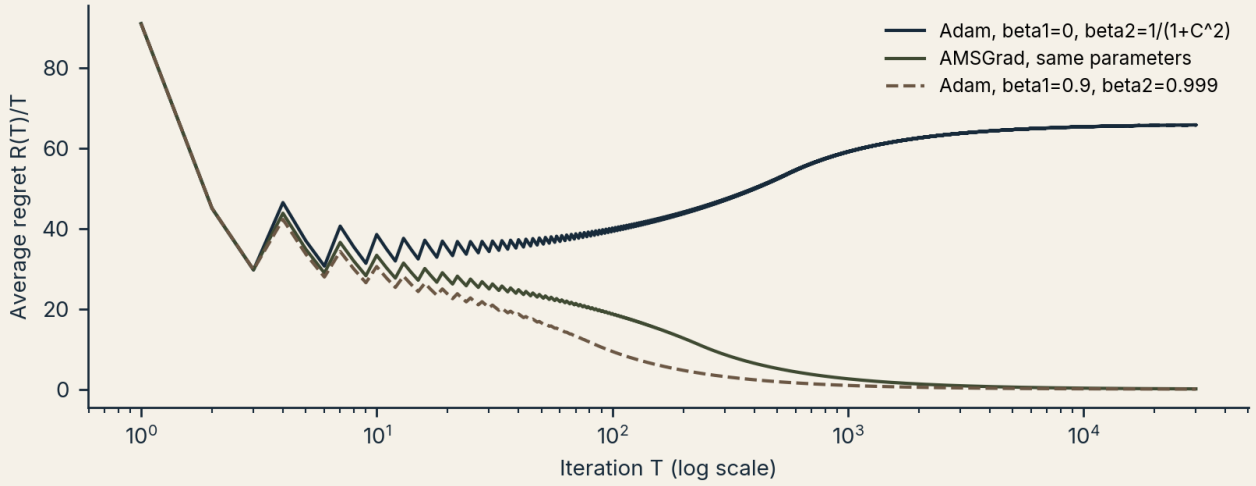
8 Contradictions and Edge Cases

CRITIQUE The conditions under which the results fail are informative. This section combines direct reasoning and a reviewer simulation.

Counterexample to convergence. Following Reddi, Kale and Kumar (2018), let $\mathcal{X} = [-1, 1]$ and let $f_t(x) = Cx$ if $t \bmod 3 = 1$ and $f_t(x) = -x$ otherwise, with $C = 101$. Over each cycle of three steps the cumulative loss is $Cx - x - x = (C-2)x$, which is minimised on $[-1, 1]$ at $x^* = -1$ with per-cycle loss $-(C-2) = -99$. A method that settles at $x = +1$ has per-cycle loss $C - 2 = 99$, so the average regret per step tends to $2(C-2)/3 = 66$. The mechanism is that the rare large gradient $+C$ is down-weighted by the exponential average faster than it is balanced by the two small opposing gradients, so the effective step on the large gradient shrinks and the iterate drifts the wrong way.

I simulated this problem with the update of Equations (1)–(4), projection on $[-1, 1]$, $\alpha_t = 0.1/\sqrt{t}$ and horizon $T = 30{,}000$. With $\beta_1 = 0$ and $\beta_2 = 1/(1 + C^2)$, the iterate sits at $+1$ in the last 3000 steps and the average regret is 65.7, close to the theoretical value of 66. Keeping the running maximum of $\hat{v}_t$ (AMSGrad) under the same parameters drives the iterate to -1 with average regret 0.09. With the defaults $\beta_1 = 0.9$, $\beta_2 = 0.999$ the failure does not appear: the iterate converges to -1 and the average regret is 0.03. Figure 2 shows the three cases.

Figure 2. Average regret on the three-step counterexample, $C = 101$



Source: generated by the reviewer; online convex problem of Reddi, Kale and Kumar (2018), $\mathcal{X} = [-1, 1]$, $T = 30{,}000$, $\alpha_t = 0.1/\sqrt{t}$, projection onto $\mathcal{X}$. Average regret is cumulative regret divided by the number of iterations, measured against $x^* = -1$.

EVIDENCE Adam with β_2 chosen small fails, as the theory of the counterexample predicts. The failure disappears at the default β_2, since a window of 1000 steps retains the large gradient long enough. INFERENCE The theorem fails for some parameter choices, but the defaults protect against this particular example. This is consistent with the observed empirical success and illustrates why the theorem and practice can diverge without either being refuted by the other.

Small gradients and the role of epsilon. The factor $g/(|g| + \varepsilon)$ equals one half when $|g| = \varepsilon = 10^{-8}$ and tends to g/ε below it. For $|g| \ll \varepsilon$ the update becomes SGD with learning rate $\alpha/\varepsilon = 10^5$ at the default values, which is enormous. Scale invariance holds only for gradient magnitudes much larger than ε, which is why networks with vanishing gradients and mixed precision arithmetic need a larger ε in practice. Reviewer calculation.

Heavy tails. A gradient of size k times the typical scale raises v_t by $(1 - \beta_2)k^2$ times the typical second moment and leaves the window only after about $1/(1 - \beta_2) = 1000$ iterations. At the default, $k = 10$ adds 0.1 of the typical second moment, a small effect; $k = 100$ adds 10, so $\sqrt{v_t}$ rises about threefold and the step is cut by roughly two thirds for a thousand iterations. Reviewer calculation. For financial data with jump risk, β_2 therefore controls the recovery time after a shock.

Non-stationarity. The bias correction in Equation (7) holds up to the residual ζ. If the gradient scale changes by a factor of ten across regimes, ζ is not small until the old regime has left the window, and the normalised step is wrong for roughly $1/(1 - \beta_2)$ steps. This is the opposite of the claim that Adam handles non-stationary objectives without qualification.

Alternative mechanism. Because the normalised step resembles sign descent, much of Adam's speed may arise from normalising coordinate scales, not from the second-moment estimate itself. Experiments in later work comparing sign-based updates with Adam are consistent with this reading (Balles and Hennig, 2018).

9 Further Literature

Table 4. **Literature connections**

Paper	Shared idea	Relationship	Why it matters
Zinkevich (2003)	Online convex programming and regret	Preceded; supplies the framework of Theorem 4.1	Defines the setting in which the guarantee is stated
Duchi, Hazan and Singer (2011)	Per-coordinate adaptive steps from accumulated squared gradients	Preceded; special case of Adam	Gives the sparse-data regret bound that Adam imitates
Tieleman and Hinton (2012)	Exponential average of squared gradients	Preceded; Adam adds momentum and correction	Source of the second-moment design
Reddi, Kale and Kumar (2018)	Non-convergence of Adam	Contradicts Theorem 4.1; proposes AMSGrad	Identifies the invalid proof step
Wilson et al. (2017)	Generalisation of adaptive methods	Contradicts the practical superiority claim for test error	Shows tuned SGD can generalise better
Loshchilov and Hutter (2019)	Decoupled weight decay	Improved Adam in practice	Corrects the interaction between L2 regularisation and adaptive scaling
Defossez et al. (2020)	Simple convergence proof for Adam and AdaGrad	Repaired the theory under different assumptions	Shows what the exponential average needs
Liu et al. (2020)	Variance of the adaptive learning rate early in training	Extended; proposes rectified Adam	Explains the need for warm-up
Gu, Kelly and Xiu (2020)	Neural networks for asset pricing	Applied; trained with Adam	Links the algorithm to quantitative finance

Source: reviewer's literature mapping; relationships stated as understood by the reviewer from the cited papers.

10 Research Gaps

10.1 Paper-level gaps

- **Gap:** A convergence guarantee for the algorithm actually used, with constant α and β_1. *Why it matters:* Users apply the guarantee informally. *Existing limitation:* The proof needs decaying parameters and a monotonicity that fails. *Way forward:* Prove convergence under conditions on β_2 relative to the horizon.
- **Gap:** Attribution of Adam's gains to momentum, scaling or correction. *Why it matters:* Guides simplification. *Existing limitation:* No factorial experiment. *Way forward:* Replicate on several architectures with matched tuning.

10.2 Literature-level gaps

- **Gap:** Why adaptive methods generalise differently from SGD. *Why it matters:* Training speed is not the objective. *Existing limitation:* Evidence is mixed across tasks. *Way forward:* Theoretical models of implicit bias for normalised updates.
- **Gap:** Optimisers under non-stationary, heavy-tailed data such as financial returns. *Why it matters:* Gradient moments are not stable. *Existing limitation:* Benchmarks are image and text. *Way forward:* Systematic study on economic datasets.

10.3 Methodological gaps

- **Gap:** Optimiser comparisons with controlled tuning budgets and many seeds. *Why it matters:* Rankings depend on the budget. *Existing limitation:* Original figures show single curves. *Way forward:* Report distributions of outcomes over seeds and tuning draws, following Choi et al. (2019).

11 Research Opportunities

Idea 1. Optimiser choice and out-of-sample performance of neural asset pricing models

RESEARCH QUESTION. Does the choice among Adam, AMSGrad and SGD with momentum change the out-of-sample Sharpe ratio and pricing error of neural network return predictors, after matching training budgets?

MOTIVATION. Financial return data have low signal to noise, so optimisation noise and implicit regularisation may matter as much as architecture.

HYPOTHESIS. Adam attains lower in-sample loss but not better out-of-sample performance than tuned SGD, and the gap grows with the share of noise in the target.

ECONOMIC MECHANISM. Implicit bias of normalised updates towards high-variance directions can fit noise factors with little pricing content.

LITERATURE CONNECTION. Gu, Kelly and Xiu (2020); Wilson et al. (2017); this paper.

DATA. Public factor and characteristic data such as the Jensen, Kelly and Pedersen global factor database, with Fama and French portfolios as a check.

VARIABLES. Monthly excess returns, firm characteristics, optimiser, learning rate, seed.

METHOD. Rolling-window estimation, ensembles over seeds, comparison of out-of-sample R^2 and long-short Sharpe.

IDENTIFICATION STRATEGY. Randomise optimiser and seed with architecture, data and training budget fixed; test differences with block bootstrap.

EXPECTED CONTRIBUTION. Quantifies whether optimiser tuning is a hidden source of reported predictability.

MAIN RISK. Differences may be smaller than seed noise.

Idea 2. Regret of adaptive optimisers on regime-switching gradients

RESEARCH QUESTION. How long does Adam take to recover its normalisation after a volatility regime change, as a function of β_2?

MOTIVATION. Equation (7) shows the residual ζ vanishes only after the window turns over.

HYPOTHESIS. Recovery time scales as $1/(1-\beta_2)$ times the log of the scale ratio between regimes.

ECONOMIC MECHANISM. Exponential averaging of squared gradients is the same filter as exponentially weighted volatility estimates in risk management.

LITERATURE CONNECTION. Reddi, Kale and Kumar (2018); RiskMetrics-type volatility filters.

DATA. Simulated Markov-switching gradient processes; then daily returns of equity indices for an online regression.

VARIABLES. Regime scale ratio, β_2, tracking error of the step normalisation.

METHOD. Analytical derivation for a two-state process, then simulation.

IDENTIFICATION STRATEGY. Controlled simulation where the true regime is known.

EXPECTED CONTRIBUTION. A practical rule for choosing β_2 under non-stationarity.

MAIN RISK. Results may be a restatement of known filter theory.

Idea 3. Adam for online portfolio selection with regret comparison

RESEARCH QUESTION. Does an Adam-type update for portfolio weights achieve lower regret against the best constant rebalanced portfolio than exponentiated gradient and the universal portfolio on real data?

MOTIVATION. Online portfolio selection is an online convex problem, the setting of Theorem 4.1, with bounded gradients on the simplex.

HYPOTHESIS. Per-coordinate scaling helps when asset volatilities differ greatly.

ECONOMIC MECHANISM. Normalising by gradient magnitude equalises learning across assets with different risk.

LITERATURE CONNECTION. Cover (1991); Helmbold et al. (1998); Zinkevich (2003).

DATA. Daily returns of the Fama and French 49 industry portfolios, public.

VARIABLES. Weights on the simplex, wealth, regret.

METHOD. Projected Adam on log-wealth, versus exponentiated gradient and online Newton step.

IDENTIFICATION STRATEGY. Out-of-sample sequential evaluation with transaction costs.

EXPECTED CONTRIBUTION. A test of the adaptive-regret theory on a real online convex problem.

MAIN RISK. Transaction costs may dominate algorithmic differences.

Idea 4. Measuring the sparsity term of the regret bound

RESEARCH QUESTION. How large is $\sum_i \|g_{1:T,i}\|_2$ relative to its worst case $d\,G_\infty\sqrt{T}$ on real tasks, and does it predict Adam's advantage over SGD?

MOTIVATION. The paper's claim of adaptive gain depends on this ratio but never measures it.

HYPOTHESIS. The ratio is small only for text-like sparse features and close to one for dense networks.

ECONOMIC MECHANISM. Not applicable; a measurement of the statistical structure of gradients.

LITERATURE CONNECTION. Duchi, Hazan and Singer (2011); this paper.

DATA. Gradient logs from public benchmark training runs.

VARIABLES. Gradient norms per coordinate, ratio, speed advantage.

METHOD. Cross-task regression of speed advantage on the ratio.

IDENTIFICATION STRATEGY. Variation across tasks with controlled training budgets.

EXPECTED CONTRIBUTION. Tests whether the theory's key quantity has empirical content.

MAIN RISK. Gradient logging at scale is expensive.

Idea 5. Optimiser-induced bias in calibrating stochastic volatility surrogates

RESEARCH QUESTION. Do neural surrogates for option pricing calibrated with Adam produce systematically different parameter errors than with second-order methods in the tails of the surface?

MOTIVATION. Calibration errors in deep wings translate directly into hedging error.

HYPOTHESIS. Adam leaves larger errors in low-gradient regions where ε and normalisation dominate.

ECONOMIC MECHANISM. Small gradients in the wings make the update behave like scaled sign descent.

LITERATURE CONNECTION. Neural calibration literature in derivatives pricing; this paper's treatment of ε.

DATA. Simulated Heston prices, then public index option data where available.

VARIABLES. Pricing error by moneyness and maturity, optimiser, ε.

METHOD. Controlled simulation with known true parameters.

IDENTIFICATION STRATEGY. Ground-truth parameters in simulation.

EXPECTED CONTRIBUTION. Practical guidance on optimiser choice in derivatives calibration.

MAIN RISK. Differences may vanish with a larger ε or better scaling.

11.1 Ranking

Table 5. **Ranking of research ideas (scores from 1 to 5, assigned by the reviewer)**

Idea	Novelty	Feasibility	Data	Method	Importance	Contribution	Total
1. Optimiser and asset pricing	4	4	4	4	5	4	25
2. Regime-switching recovery	4	5	5	4	3	3	24
3. Online portfolio selection	3	4	5	4	3	3	22
4. Sparsity term measurement	4	3	3	4	3	4	21
5. Calibration surrogates	3	3	3	3	4	3	19

Source: reviewer's subjective scoring.

The strongest three are Ideas 1, 2 and 3. Idea 1 has the highest importance, since it tests whether a hidden tuning choice influences published return predictability, and the data are public. Idea 2 is the most tractable: it has a closed-form core and a clean simulation design. Idea 3 connects the theorem directly to a real online convex problem with free data.

12 Learning Framework

12.1 Prerequisite concepts

- Convexity and subgradients; online convex optimisation and regret.
- Stochastic gradient descent, momentum and Nesterov acceleration.
- Exponential moving averages and their bias at initialisation.
- Young's inequality and telescoping sums for regret proofs.

12.2 Core concepts learned

- Per-coordinate normalisation by a running second moment.
- Zero-initialisation bias and its correction.
- Regret as a performance criterion for sequential decisions.
- The gap between a convex guarantee and non-convex practice.

12.3 Important equations

Equations (1) to (4) define the algorithm. Equation (7) justifies bias correction. Equation (5) bounds the step. Equation (8) defines regret. Equation (12) links Adam to AdaGrad.

12.4 Methods learned

Regret analysis of a projected adaptive method; bias correction of exponential averages; use of a counterexample to test a theorem numerically.

12.5 Questions to test understanding

COMPREHENSION

1. Why is $v_1 = 0.001\, g_1^2$ at the first step with $\beta_2 = 0.999$, and what does the correction do?
2. What is the largest possible step of Adam with the defaults, and in which case does it occur?
3. How is AdaGrad obtained from Adam?

ANALYSIS

1. Which step of the regret proof fails, and why does a running maximum of $\hat{v}_t$ repair it?
2. Why does the counterexample fail at $\beta_2 = 0.999$ but succeed at $\beta_2 = 1/(1 + C^2)$?
3. When does ε break scale invariance?

RESEARCH

1. Design an experiment separating the role of momentum, scaling and correction.
2. How would heavy-tailed gradients change the effective memory of $\hat{v}_t$?
3. What guarantee could be proved for constant α and β_1?

12.6 Papers to read next

Reddi, Kale and Kumar (2018); Wilson et al. (2017); Defossez et al. (2020); Loshchilov and Hutter (2019); Duchi, Hazan and Singer (2011).

13 Conclusion

13.1 One-sentence thesis

Adam combines momentum, per-coordinate normalisation and bias correction into a robust default optimiser whose empirical value is well supported, while its accompanying convergence guarantee is invalid as stated.

13.2 Three most important insights

- Bias correction is necessary for the default $\beta_2 = 0.999$ to be usable.
- The normalised step is bounded and, apart from ε, scale invariant.
- A proof can be wrong while the algorithm works: the theorem fails for some parameters but not at the defaults.

13.3 Strongest evidence

The variational autoencoder sweep with and without bias correction, which changes one component only, and the analytical Equation (7).

13.4 Biggest weakness

The telescoping step of the regret proof, which requires monotonicity that exponential averaging does not provide.

13.5 Most important research gap

A convergence theory for the constant-parameter algorithm actually used, together with an explanation of its generalisation behaviour.

13.6 Best research opportunity

Testing whether optimiser choice alters out-of-sample performance of neural return predictors under matched budgets.

13.7 Three papers to read next

Reddi, Kale and Kumar (2018); Wilson et al. (2017); Gu, Kelly and Xiu (2020).

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